

1.5 hours on examples

Problem 2 (98AMO1)

We let the number of 6s be x and number of 1s be $999 - x$.

$$\sum |a_i - b_i| = 6x + (999 - x)$$

Note that $1 \equiv \sum (a_i + b_i) \equiv \sum a_i - \sum b_i \equiv \sum |a_i - b_i| \equiv 1 + x \pmod{2}$. Therefore, we must have an odd number for x and an even number of 6s.

$$\underline{6x + (999 - x) \equiv 9 - 5x \equiv 9 \pmod{10}}$$

Problem 4 (NIMO237)

Note that by Fermat's Little Theorem, $a^p \equiv a \pmod{p}$ for all $1 \leq a \leq 2015$. Therefore,

$$\left\lfloor \frac{a^p}{p} \right\rfloor = \frac{a^p - a}{p}$$

$$\left\lfloor \frac{1^p}{p} \right\rfloor + \left\lfloor \frac{2^p}{p} \right\rfloor + \dots + \left\lfloor \frac{2015^p}{p} \right\rfloor = \frac{(1^{2017} - 1) + (2^{2017} - 2) + (3^{2017} - 3) \dots + (2015^{2017} - 2015)}{2017}$$

Note that $\sum_{i=1}^{2015} i = 1008 \cdot 2015 \pmod{2017^2}$.

We now calculate,

$$\sum_{i=1}^{2016} i^{2017} \equiv \sum_{i=1}^{1008} i^{2017} + (2017 - i)^{2017} \equiv 0 \pmod{2017^2}$$

Therefore,

$$\begin{aligned} \sum_{i=1}^{2015} i^{2017} - \sum_{i=1}^{2015} i &\equiv -2016^{2017} - 1008 \cdot 2015 \pmod{2017^2} \\ &\equiv (2017 - 1)^{2017} - 1008 \cdot 2015 \pmod{2017^2} \\ &\equiv 1 - 1008 \cdot 2015 \equiv 2017(1010) \pmod{2017^2} \end{aligned}$$

Therefore, $\left\lfloor \frac{1^p}{p} \right\rfloor + \left\lfloor \frac{2^p}{p} \right\rfloor + \dots + \left\lfloor \frac{2015^p}{p} \right\rfloor \equiv 1010 \pmod{2017}$.

Problem 5 (NIMO157)

Let,

$$f(x) = x^6 + x^5 + x^3 + 1 = (x + 1)(x^2 + 1)(x^3 - x + 1)$$

Plug in $x = 20$ to get $67208001 = (21)(401)(7981) = (3)(7)(23)(347)(401)$

The answer is hence 781.

3 hours; 10 clubs

Do you know what score the following proof will get on the contest? Is there not enough info?

Problem 7 (09IMO1)

We define $a_{k+b} = a_b$.

For the sake of contradiction, let,

$$n \mid a_1(a_2 - 1);$$

$$n \mid a_2(a_3 - 1);$$

...

$$n \mid a_{k-1}(a_k - 1);$$

$$n \mid a_k(a_1 - 1);$$

Claim: If $\gcd(a_i, n) > 1$ for some i , we have $n \mid a_i$ for all i . Similarly, if $\gcd(a_i - 1, n) > 1$ for some i , we have $n \mid a_i - 1$ over all i . Exactly one of these conditions are met.

Proof: We use the fact that we cannot have both $\gcd(a_i, n) > 1$ and $\gcd(a_i - 1, n) > 1$.

WLOG, if $\gcd(a_k, n) > 1$, then we use $n \mid a_{k-1}(a_k - 1)$ to get $n \mid a_{k-1}$, repeating the process until we get $n \mid a_i$ for all i .

If neither condition is met, then nothing can be divisible by n . It is impossible for both to be met because we cannot have both $\gcd(a_i, n) > 1$ and $\gcd(a_i - 1, n) > 1$.

Using the claim, we either have $n \mid a_1, a_2, \dots, a_k$ (exclusive) or $n \mid a_1 - 1, a_2 - 1, \dots, a_k - 1$. It is impossible to find at least k multiples of n on $[1, n]$ for $k \geq 2$, nor is it possible to find at least k numbers that are one more than a multiple of n .

Problem 8 (20IBERO2)

We claim the answer is every number apart from the powers of 2 greater than 2.

For the sake of contradiction, we let $n = 2^k$ for some $k \geq 1$ have $T_m \leq m$.

$$v_2(1 + 2 + \dots + T_{m+1}) = v_2\left(\frac{(T_m)(T_m + 1)}{2}\right) = v_2(T_m) + v_2(T_m + 1) - 1 \geq k$$

At most one of $v_2(T_m)$ and $v_2(T_m + 1)$ can be greater than 1 and neither of them can be more than k , so this is impossible.

We prove that all other n work. Note that,

$$\begin{aligned} x(x+1) &\equiv 0 \pmod{2n} \\ \Leftrightarrow (2n-x-1)(2n-x) &\equiv 0 \pmod{2n} \end{aligned}$$

Therefore, if there's any solution to $x(x+1) \equiv 0 \pmod{2n}$ for $1 \leq x \leq 2n-2$, we are done.

Since there must be another prime factor of $2n$, we can write $2n = ab$ for $(a, b) = 1$ and $a, b > 1$. By CRT, we solve for solutions to,

$$x(x+1) \equiv 0 \pmod{a};$$

Of which we have $x \equiv 0, a-1 \pmod{a}$.

And multiply that with the number of solutions to,

$$x(x+1) \equiv 0 \pmod{b};$$

Of which we have $x \equiv 0, b-1 \pmod{b}$.

Therefore, we have at least 4 solutions, two of which must be different from 0 and $2n-1 \pmod{2n}$ by Pigeonhole. We are hence done.

4 hours; 20 clubs

Problem 10 (PRIMES 2019 entrance exam)

We claim the answer is all number of the form $2p$ and p^k except 2.

To construct the $2p$, we can have,

$$\begin{aligned} p &\equiv 0 \pmod{\text{lcm}(1, 2, 3, \dots, 2p-1)}; \\ p &\equiv p \pmod{2p} \end{aligned}$$

This way p divides all the numbers from 1 ... $2p-1$ but it fails at $2p$.

To construct p^k , we can have,

$$\begin{aligned} p &\equiv 0 \pmod{\text{lcm}(1, 2, 3, \dots, p^k-1)}; \\ p &\equiv p^{k-1} \pmod{p^k} \end{aligned}$$

Since $p^k \neq 2$, the last quantity cannot be $\equiv 0, -1 \pmod{p^k}$.

We now prove that all other numbers don't work.

2 obviously doesn't work since 2 divides one of $n, n + 1$.

All numbers of the form $n = ab$ for $\gcd(a, b) = 1$ and $a > 2; b > 2$ don't work. Since,

$$2a < n; 2b < n$$

We take,

$$n \equiv 0, -1 \pmod{2a}; n \equiv 0, -1 \pmod{2b}$$

Since n is either even or odd, we cannot have $n \equiv 0 \pmod{2a}; n \equiv -1 \pmod{2b}$ or vice versa. Therefore,

$$n \equiv 0, -1 \pmod{2ab} \Rightarrow n \equiv 0, -1 \pmod{ab}$$

Finally, all numbers of the form $n = 2p^k$ for $k > 1$ don't work.

$$n \equiv 0, -1 \pmod{p^k}; n \equiv 0, -1 \pmod{2p^{k-1}} \Leftrightarrow n \equiv 0, -1 \pmod{2p^k}$$

Once again, we have a contradiction.

Problem 11 (USEMO 2019/4)

Note that if we plug in $n = p(p - 1)$ into the expression, we have,

$$\begin{aligned} & 1^{p(p-1)} + 2^{(p-1)(p-1)+(p-2)} + \dots + p^{(p-1)(p-1)} \\ & + (p+1)^{(p-2)(p-1)+(p-2)} + \dots + 2p^{(p-3)(p-1)+(p-2)} \\ & \dots \\ & + (p(p-2)+1)^{(p-1)+1} + \dots + ((p-1)p)^1 \\ & \equiv 1^0 + 2^{p-2} + \dots + (p-1)^1 + p^0 \\ & \quad + 1^1 + 2^0 + \dots + (p-1)^2 + p^1 \\ & \dots \\ & + 1^{p-2} + 2^{p-3} + \dots + (p-1)^0 + p^{p-2} \\ & \equiv (1^0 + 1^1 + \dots + 1^{p-2}) + \sum_{i=2}^{p-1} \frac{i^{p-1} - 1}{i - 1} \end{aligned}$$

Since all numbers from 1 to $p - 2$ have a non-zero modular inverse mod p ,

$$\equiv (p-1) + \sum_{i=2}^{p-1} \frac{1-1}{i-1} \equiv p-1$$

Now, note that if we plug in $n = k(p)(p - 1)$, we'll get the expression $\equiv n - k \pmod{p}$. Therefore, we can attain all the mod p 's so we're done.

6 hours; 30 clubs

Misplaced problem? I'm not sure how this is Diophantine equations.

Problem 16 (Z57CC72B / Wolstenholme's Theorem)

Being mindful that $p \geq 5$ is a prime,

$$\begin{aligned} p^2 &\mid (p-1)! \left(\frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{p-1} \right) \\ \Leftrightarrow p^2 &\mid 2(p-1)! \left(\frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{p-1} \right) \\ \Leftrightarrow p^2 &\mid p! \left(\sum_{i=1}^{p-1} \frac{1}{p(p-i)} \right) \end{aligned}$$

Negatives can go away,

$$\Leftrightarrow p \mid \left(\sum_{i=1}^{p-1} i^{-2} \right)$$

The modular inverses must all be distinct, so we have,

$$\Leftrightarrow p \mid \left(\sum_{i=1}^{p-1} i^2 \right) = \frac{(p-1)(p)(2p-1)}{6}$$

This is true for $p \geq 5$.

Problem 18 (JMO 2011)

We claim the only solution is $n = 1$, which obviously works $45^2 = 2025$.

$n = 0$ does not work. For $n > 1$, we take mod 4,

$$2^n + 12^n + 2011^n \equiv (-1)^n \pmod{4}$$

Since squares are either 0, 1 mod 4, we have $2 \mid n$. We write $n = 2k$ for some integer $k \geq 1$. Note that,

$$2011^{2k} = (2011^k)^2 < 2011^{2k} + 12^{2k} + 2^{2k} < (2011^k + 1)^2 = 2011^{2k} + 2 \cdot 2011^k + 1$$

This is true since $144^k + 4^k < 2011^k + 2011^k + 1$ for $k \geq 1$.

Problem 20 (Shortlist 2002)

We claim the answer is 4. To achieve this, we have,

$$(2002^{667})^3 + (2002^{667})^3 + (10 \cdot 2002^{667})^3 + (10 \cdot 2002^{667})^3 = 2002^{2002}$$

We take mod 9,

$$\sum_{k=1}^t x_k^3 \equiv 2002^{2002} \equiv 4^4 \equiv 4 \pmod{9}$$

However, cubes are $-1, 0, 1 \pmod{9}$, so we need to add at least 4 cubes to get 4 mod 9.

Problem 21 (Pen H61)

We claim the only solutions are $(x, y) = (4, 1)$. Obviously, no solution works for $x = 0, 1, 2, 3$. We prove by contradiction that no integer y works for $x > 4$.

No solution works for $y = 0$, so we take mod 11 to get,

$$2^x \equiv 11^y + 5 \equiv 5 \pmod{11} \Leftrightarrow x \equiv 4 \pmod{10}$$

Therefore, x is even, so we write $x = 2k$.

Taking mod 32,

$$\begin{aligned} 2^x &\equiv 0 \equiv 11^y + 5 \pmod{32} \\ &\Leftrightarrow 27 \equiv 11^y \pmod{32} \end{aligned}$$

We see that,

$$\begin{aligned} 11^0 &\equiv 1 \pmod{32}; 11^1 \equiv 11 \pmod{32}; 11^2 \equiv 25 \pmod{32}; 11^3 \equiv 19 \pmod{32}; \\ 11^4 &\equiv 17 \pmod{32}; 11^5 \equiv 27 \pmod{32}; 11^6 \equiv 9 \pmod{32}; 11^7 \equiv 3 \pmod{32}; 11^8 \equiv 1 \pmod{32} \end{aligned}$$

Since we get that, $\text{ord}_{32}(11) = 8$,

$$27 \equiv 11^y \pmod{32} \Leftrightarrow y \equiv 5 \pmod{8}$$

Now, taking mod 17,

$$2^x = 4^k \equiv 11^y + 5 \pmod{17}$$

Note that, $4^k \equiv 1, 4, 16, 13 \pmod{17}$. Since $y \equiv 5 \pmod{8}$, $11^y \equiv 11^5, 11^{13} \equiv 10, 7 \pmod{17}$

None of these pairs will work for $4^k \equiv 11^y + 5 \pmod{17}$ so we are done.

9 hours; 38 clubs

Problem 22 (TSTST 2017/4)

We claim the only solutions are $(a, b, c, n) = (1, 1, 0, 3)$ or $(2, 0, 0, 3)$, which obviously works. We prove that there exist no other solutions.

Evidently, there are no solutions for $n = 0, n = 1, n = 2$. For $n = 3$,

$$2^a + 3^b + 5^c = 6$$

If $c > 0$, too much, so $c = 0$ and we get the only solutions are the ones above.

For $n = 4$,

$$2^a + 3^b + 5^c = 24$$

If $c > 2$, too much. If $c = 1$, $2^a + 3^b = 19$ gives no solutions. If $c = 0$, $2^a + 3^b = 23$ also gives none.

For $n \geq 5$, we take modulo 120,

$$2^a + 3^b + 5^c \equiv 0 \pmod{120}$$

We look at the powers of 2 mod 120, $2^0 \equiv 1; \dots; 2^3 \equiv 8; 2^4 \equiv 16; 2^5 \equiv 32; 2^6 \equiv 64; 2^7 \equiv 8$ and it repeats 8, 16, 32, 64 from here.

For powers of 3, $3^0 \equiv 1; 3^1 \equiv 3; 3^2 \equiv 9; 3^3 \equiv 27; 3^4 \equiv 81; 3^5 \equiv 1$ so it repeats 1, 3, 9, 27, 81.

For powers of 5, $5^0 \equiv 1; 5^1 \equiv 5; 5^2 \equiv 25; 5^3 \equiv 5$ and it repeats 5, 25 from here.

We have,

$$2^n \equiv 1, 2, 4, 8, 16, 32, 64 \pmod{120}$$

$$3^n \equiv 1, 3, 9, 27, 81 \pmod{120}$$

$$5^n \equiv 1, 5, 25 \pmod{120}$$

We can quickly look through all combinations and check that none add to 0 mod 120.

Problem 23 (IMO 2006/4)

We claim that the only answer is $(x, y) = (0, 2)$, which obviously works.

$$1 + 2^x + 2^{2x+1} = y^2$$

$$\Leftrightarrow 2^x(2^{x+1} + 1) = (y - 1)(y + 1)$$

If y is even, the RHS is odd, so we write $y = 2k + 1$.

$$\Leftrightarrow 2^x(2^{x+1} + 1) = (2k)(2k + 2)$$

$$\Leftrightarrow 2^{x-2}(2^{x+1} + 1) = k(k + 1)$$

$$\Leftrightarrow 2^{2x-1} + 2^{x-2} = k^2 + k$$

Case 1: $2^{x-2} \mid k$.

Plugging in $k = 0$, $2^{2x-1} + 2^{x-2} = 0$, impossible.

Plugging in $k = 2^{x-2}$, $2^{2x-1} + 2^{x-2} = 2^{2x-4} + 2^{x-2}$. Impossible.

Plugging in $k = 2 \cdot 2^{x-2}$, $2^{2x-1} + 2^{x-2} = 2^{2x-2} + 2^{x-1} \Leftrightarrow 2^{2x-2} = 2^{x-2}$. This yields the solution above $x = 0$, $y = 2k + 1 = 2$.

Plugging in $k = 3 \cdot 2^{x-2}$, $2^{2x-1} + 2^{x-2} = 9 \cdot 2^{2x-4} + 3 \cdot 2^{x-2} > 2^{2x-1} + 2^{x-1}$. This, and any higher solutions would fail by a similar limiting argument.

Case 2: $2^{x-2} \mid k + 1$.

Plugging in $k = 2^{x-2} - 1$, $2^{2x-1} + 2^{x-2} = 2^{2x-4} - 2^{x-2}$. The RHS is obviously too low.

Plugging in $k = 2 \cdot 2^{x-2} - 1$, $2^{2x-1} + 2^{x-2} = 2^{2x-2} - 2^{x-1} \Leftrightarrow 2^{2x-2} = -2^{x-2}$. Still too low.

Plugging in $k = 3 \cdot 2^{x-2} - 1$,

$$\begin{aligned} 2^{2x-1} + 2^{x-2} &= 9 \cdot 2^{2x-4} - 3 \cdot 2^{x-2} \\ \Leftrightarrow 2^x &= 2^{x-4} \end{aligned}$$

This is impossible.

Plugging in $k = 4 \cdot 2^{x-2} - 1$,

$$2^{2x-1} + 2^{x-2} = 2^{2x} - 2^x \Leftrightarrow 5 \cdot 2^{x-2} = 2^{2x-1} \Rightarrow 2^{x+1} > 2^{2x-1}$$

The last statement is untrue for $x > 1$. For any higher, we use a similar limiting argument.

11 hours; 48 clubs

Problem 26 (AIME 2021/14)

We claim the lowest positive n is 1952286. We'll prove that this is attainable and minimal below.

$a = 1$ always works obviously.

Since $\sigma(n)$ is a multiplicative function, we have $\sigma(a^n) \equiv 1 \pmod{2021}$ for all positive integers $a > 1$ if and only if, for all primes p and integers e , $\sigma(p^{en}) \equiv 1 \pmod{2021}$.

With modulo 43,

$$\sigma(p^{en}) = 1 + p + p^2 + \dots + p^{en} = \frac{p^{en+1} - 1}{p - 1} \equiv 1 \pmod{43}$$

If $p \equiv 1 \pmod{43}$ (for example $p = 4 \cdot 43 + 1$),

$$\sigma(p^{en}) = 1 + p + p^2 + \dots + p^{en} \equiv (1)(en + 1) \equiv 1 \pmod{43}$$

We get that $43 \mid n$.

If $p \not\equiv 1 \pmod{43}$ (for example $p = 2$),

$$\sigma(p^{en}) = 1 + p + p^2 + \dots + p^{en} \equiv 1 \pmod{43}$$

$$\Leftrightarrow \frac{p(p^n - 1)}{p - 1} \equiv 0 \pmod{43}$$

Since $p - 1$ has a modular inverse mod 43,

$$\Leftrightarrow p(p^n - 1) \equiv 0 \pmod{43}$$

If $43 \mid p$, we're done. If that's not the case, $43 \mid n$.

We make a similar argument for mod 47 (our primes being $p = 6 \cdot 43 + 1$; $p = 2$ respectively) to get,

$$\text{lcm}(42, 43, 46, 47) = 1952286 \mid n$$

Problem 27 (AIME II 2022/13)

Since $x \neq \pm 1$,

$$P(x) = (1 + x^{30} + \dots + x^{2280})(1 + x^{42} + \dots + x^{2268})(1 + x^{70} + \dots + x^{2240})(1 + x^{105} + \dots + x^{2205})(x^{2310} - 1)^7$$

We want absolute value of the coefficient of x^{2022} and count how many ways we can choose the terms in each of the 11 factors. For the $(x^{2310} - 1)^7$, we must choose the -1 every time to prevent going over, so it contributes nothing.

For the other terms, since the degree term of each factor is greater than 2022, we're looking for tuples of non-negative a, b, c, d integers such that,

$$30a + 42b + 70c + 105d = 2022$$

Taking modulo 7, 5, 3, 2 respectively, we get,

$$a \equiv 3 \pmod{7}; b \equiv 1 \pmod{5}; c \equiv 0 \pmod{3}; d \equiv 0 \pmod{2}$$

We can hence write,

$$a = 7A + 3; b = 5B + 1; c = 3C; d = 2D$$

We count the number of non-negative A, B, C, D to get the answer,

$$30a + 42b + 70c + 105d = 2022$$

$$\Leftrightarrow 210(A + B + C + D) = 1890$$

By stars and bars, there are $\binom{9 + 4 - 1}{4 - 1} = \binom{12}{3} = 220$ ways.

13 hours; 54 clubs (excl. mini survey)